AWS Training - Practical Scenario:

Owner UT

Subject Docker - Lab book.

1. Docker installation
2. What is the gpg key?
3. Docker create, run, run -itd, run -it -rm, exec, exec -it start, stop, remove, pause, unpause, rm, rmi, rm -f, rmi -f, save, load, commit, build
4. Uninstall docker
5. docker services
6. Docker images - Templates (docker registry).
7. Pull the images.
8. Pull the images with the version.
9. Check the local images.
10. Run the container with 8000 port
11. Check the docker container status
12. Check the hardware utilization for docker container
13. Set the RAM memory for container
14. How to do memory swap for the container
15. How to set the limit for cpu
16. How to share the cpu core?
17. How to set the pid limit?

18. What does docker update

19. What is the docker update cmd restriction?

20. What does the docker log?

21. Stop and start the container

22. Stop the multiple container

23. Stop all the containers?

24. What is docker ps -a

25. What is the difference between a stopped and exited container?

26. Remove the particular container?

27. Do force remove the particular container

28. How to remove all the stopped/exited containers?

29. How to remove all unused resources in docker?

30. Pause and unpause the container

31. Delete the docker image

32. Docker run command

33. Docker run -it cmd?

34. How to work inside the container

35. How to take container backup as an image.

36. Move the docker image to the local server

37. Move the images from local to docker.

38. Push image from docker to docker_hub

39. What is docker tag

40. Create docker file - build an own docker image

41. How do docker containers run with the docker host OS?

42. Where the docker is sending their commands to execute?

43. What is Docker Daemon service?

44. What is Docker Registry?

45. What is the container lifecycle?

46. What is containerd?

47. What is runC?

48. Docker pull command

49. Docker image commands

50. Docker ps cmd?

51. What is docker create cmd?

52. Linux cmd to check the docker process?

53. What is docker exec cmd?

54. List of run cmd?

55. What is the docker inspect cmd?

56. Docker logs cmd?

57. docker help cmd?

58. docker image prune cmd?

59. What is a docker file?

60. docker Image Layers?

61. What is layer cache?

62. What is RUN cmd in docker file?

63. What is CMD in the docker file?

64. What is the docker history command?

65. What is docker expose?

66. What is docker COPY in file?

67. Docker WORKDIR, ENV, LABEL instructions?

68. List the Docker File structure?

69. Docker Registry - Pushing Images?

70. Docker Image tag?

71. Docker Image Lifecycle?

72. Docker Networking?

73. Docker Bridge Networks?

74. Docker Network Modes?

75. Docker bridge driver / mode?

76. Docker Default Bridge (docker0)?

77. Docker Networking commands?

78. Docker Persisting Data in Docker [backup the container data]?

79. Docker Volumes?

80. docker volume commands?

81. Docker run command options & docker volume?

82. Sharing data between multiple docker containers

83. Persisting data using bind mounting:

84. Docker Compose

85. Docker Compose Network

86. Deploy a Website on Nginx using a Dockerfile

87. How is a Docker client different from a Docker daemon? Can you share an example?

88. Can you explain what Docker networking is and which commands can create a bridge and an overlay network?

89. Explain how Docker bridge networking works.

90. What is a Dockerfile? Explain how you would write it.

91. What is Docker Compose, and how is it different from Dockerfile?

92. Why do we use volumes in Docker?

93. What are Docker bind mounts, and why do we prefer volumes over bind mounts?

94. What is Docker Swarm?

95. Can we autoscale Docker Swarm

96. Can a container restart by itself? Define its default and always policies.

97. Explain the lifecycle of Docker containers.

98. Tell me about 3 best practices to keep a Docker container safe.

99. Why do Docker containers need health checks?

100. What are dangling images in Docker, and how do you remove them?

101. What is the primary difference between Docker and Kubernetes?

102. Compare Docker Swarm with Kubernetes.

103. How does Kubernetes manage a large number of Docker containers?

104. What is a Pod in Kubernetes, and how is it different from a container?

105. How can you manage sensitive data like passwords in Docker and Kubernetes?

106. Imagine you're creating an image of a Maven-based API. You've already set up the Dockerfile with basic configurations. Now, you notice the image size is large. How would you reduce it?

107. Imagine you need to push a Docker container image to the Docker Hub with Jenkins. How would you do that?

108. Imagine you have to migrate a WordPress Docker container to a new server without losing any data. How would you do that?

109. Create custom networks (bridge, host, overlay)

110. Connect containers to specific networks

111. Expose and publish ports

112. Configure logging drivers

113. Tagging images

114. Add custom tags to images

115. Understand image versioning

116. Define multi-container applications in docker-compose.yml

117. Cleanup unused images, containers, and volumes (docker system prune)

118. Deploy services across multiple nodes.